Photometric Variations of Luminous Blue Variables in M31 & M33: Diversity in S Doradus Behaviors

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Agenda

01	Luminous Blue Variables
02	Methodology
03	Results
04	Takeaways

Luminous Blue Variables

- Also classified as Hubble-Sandage variables (Hubble & Sandage 1953) or S Dor variables (Wolf 1989)
- Have the **highest** mass loss rates of any stars (Smith et. al. 2020)
- Unstable, massive, evolved stars with irregular outbursts, leading to the formation of a "pseudo-photosphere" (Szeifert et. al. 1996)

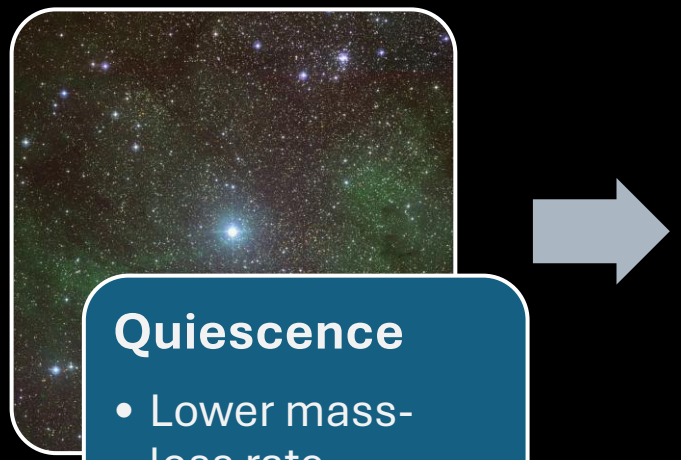


(NASA, ESA, N. Smith (U. Arizona) and J. Morse (BoldlyGo Institute)



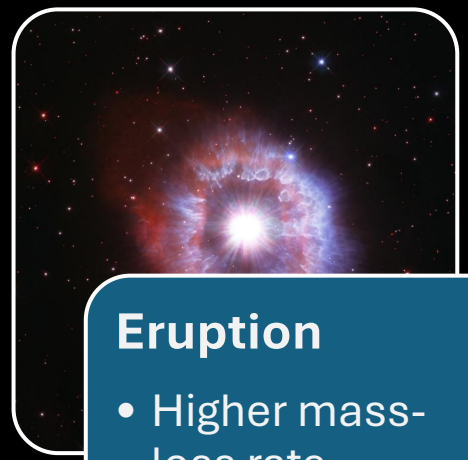
(Credit: NASA, ESA, STScI)

Traditional Interpretation



Quiescence

- Lower mass-loss rate
- "Normal" high temperature (> 15,000 K)

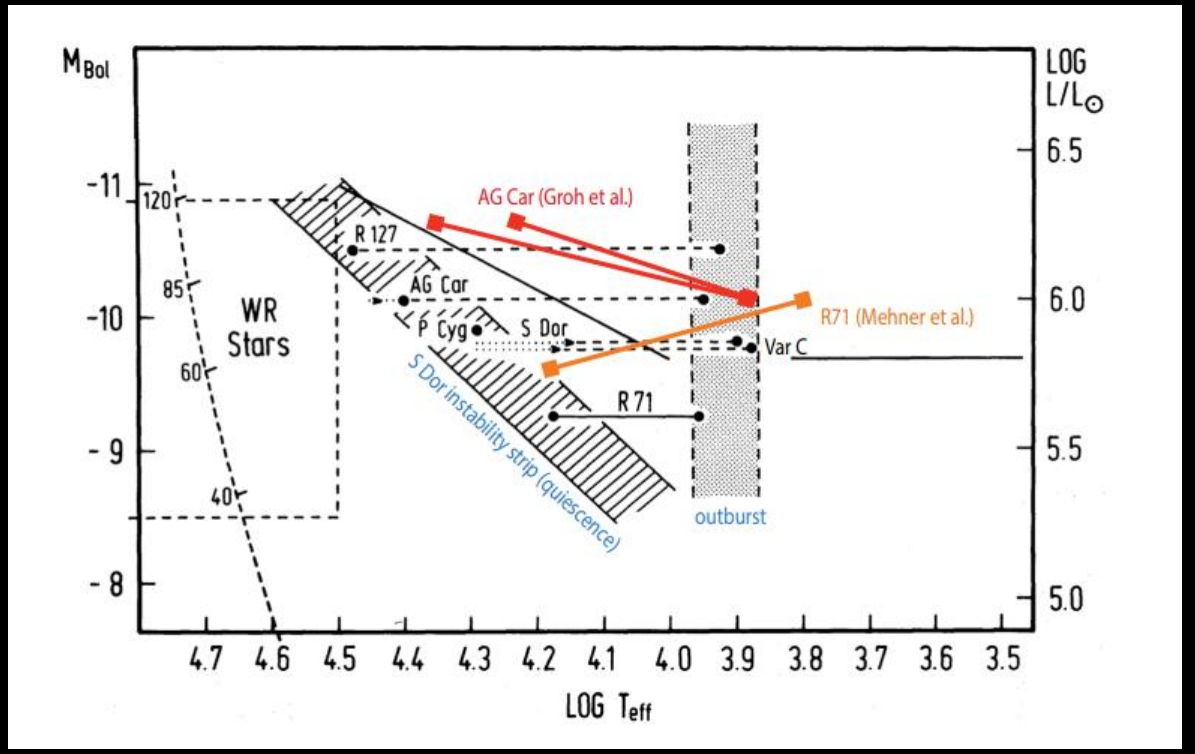


Eruption

- Higher mass-loss rate
- Cool (7000-9000 K) and dense expanding envelope

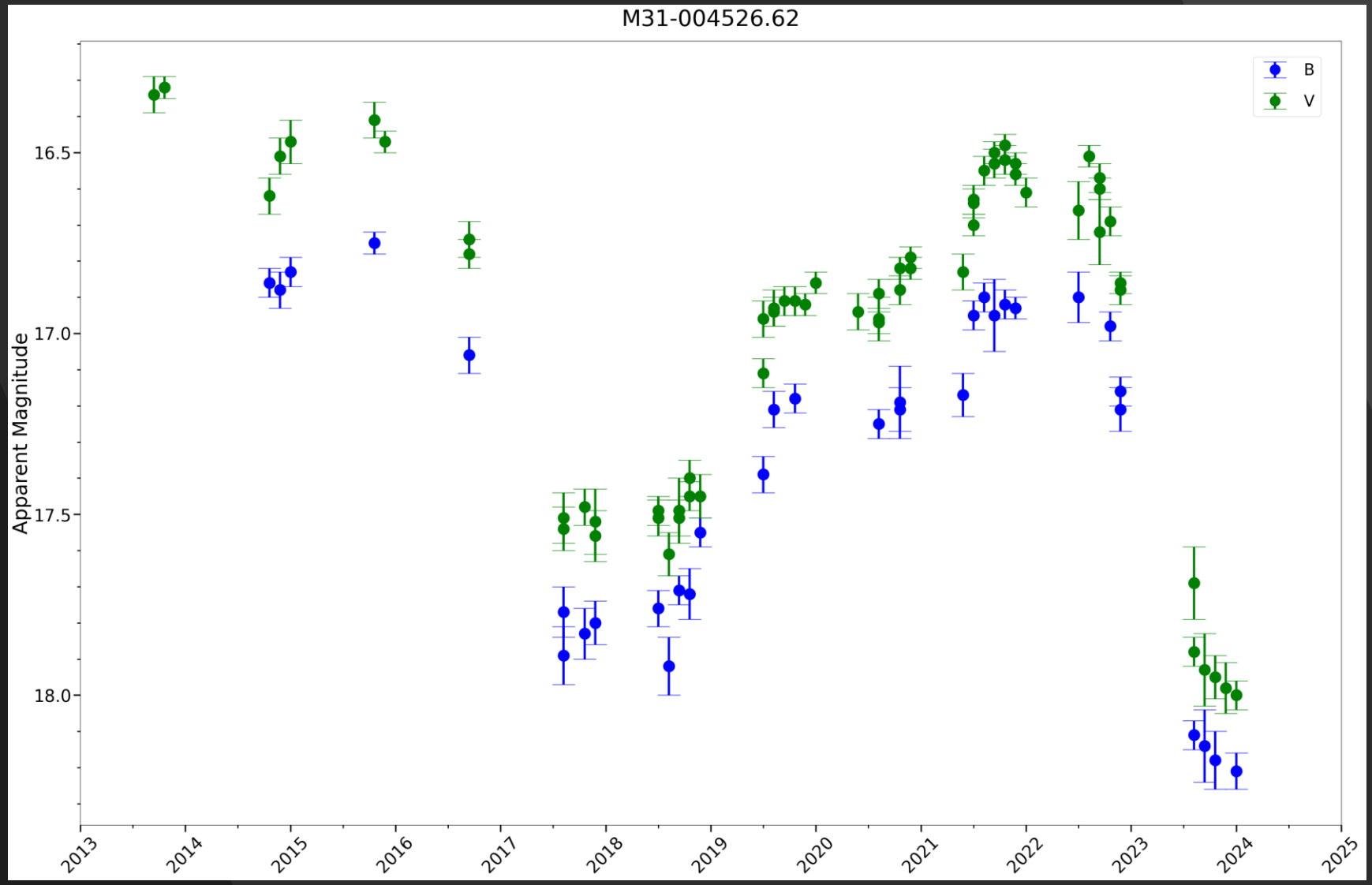
Brighten in optical wavelengths at **constant bolometric luminosity** (Szeifert et. al. 1996, see also Humphreys & Davidson 1994)

The Twist



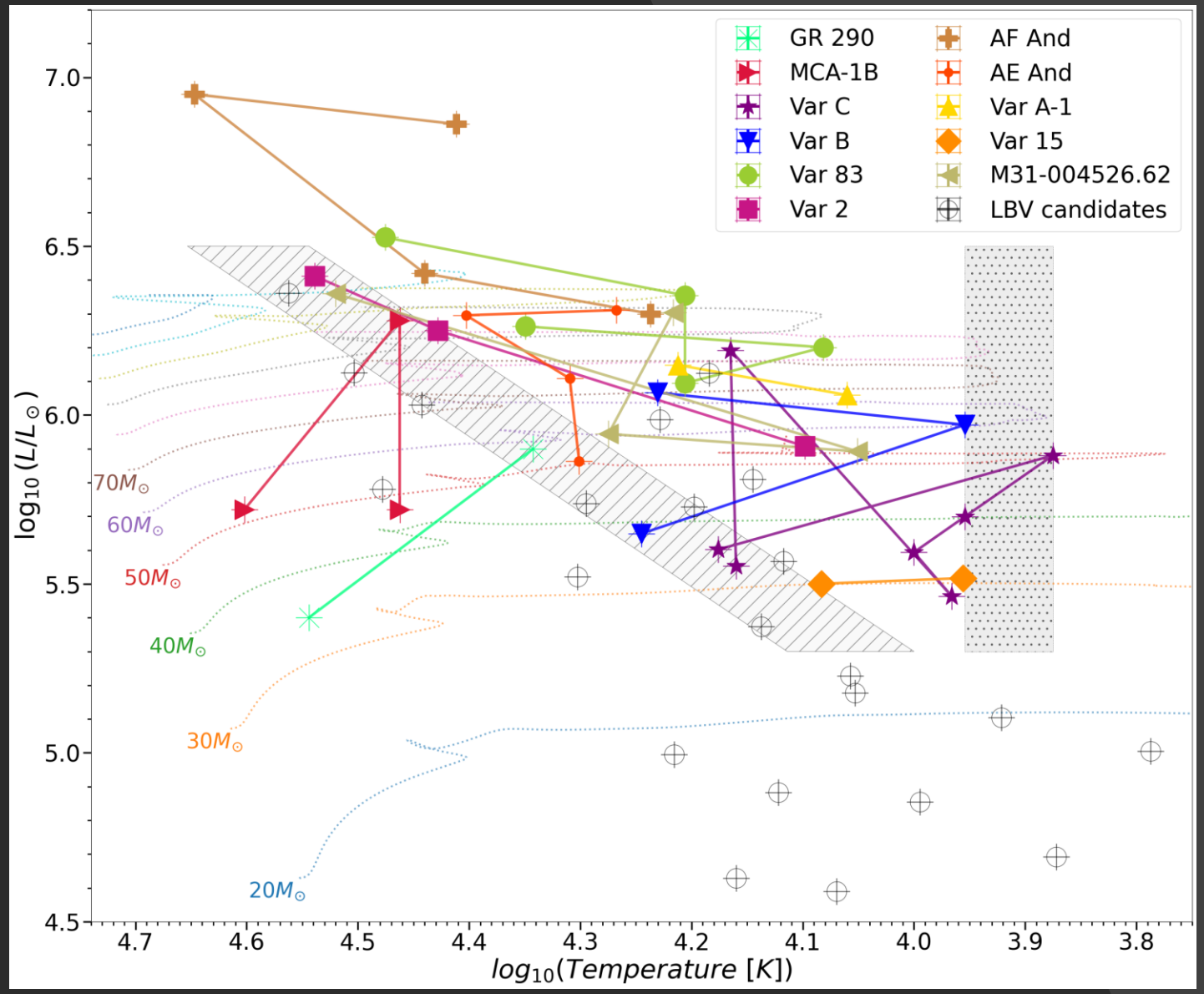
N. Smith (2026)

Example Light Curve

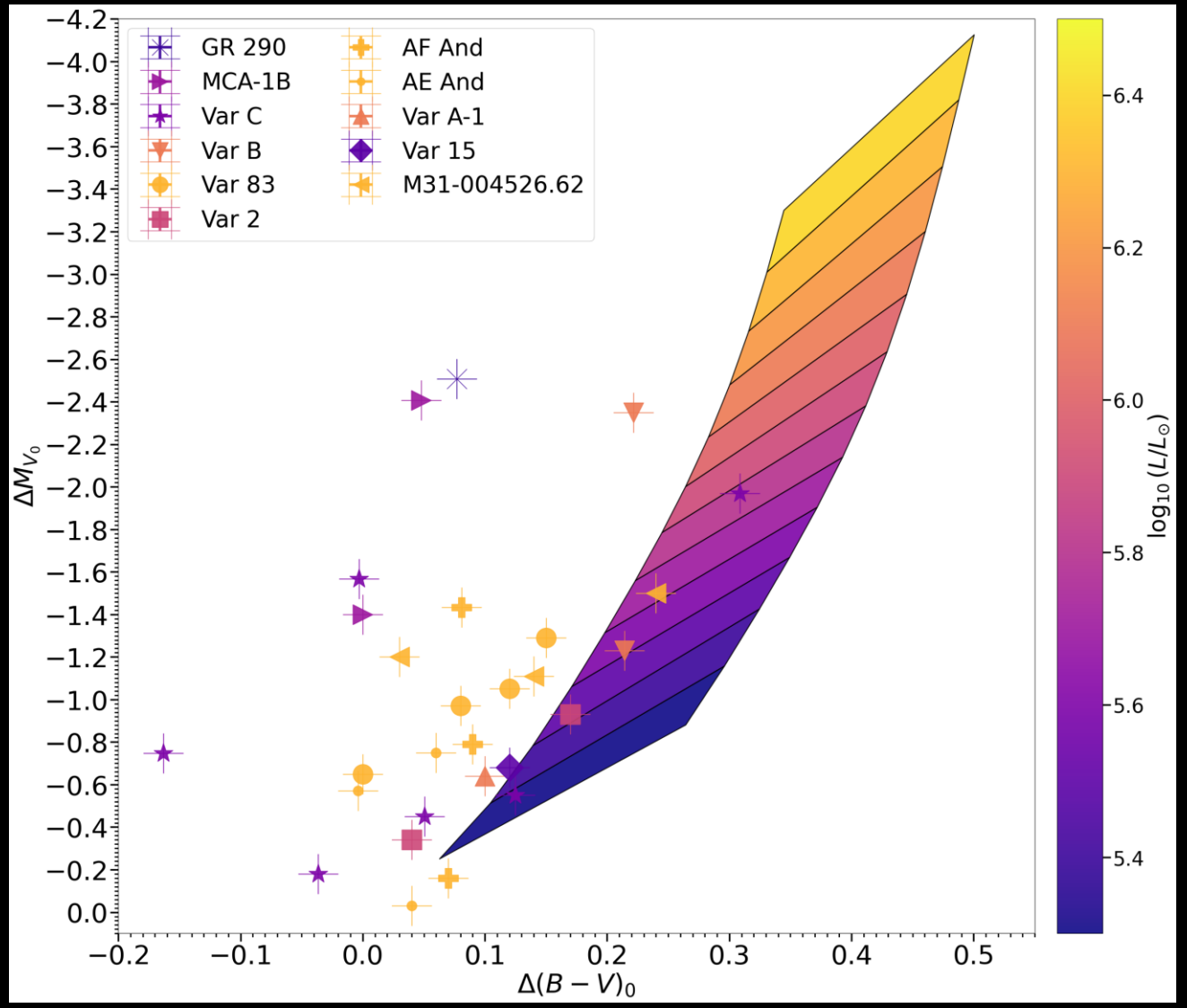


HR Diagram

- Obtain B and V photometry for 11 confirmed and 26 candidate LBVs from past literature and publicly available data.
- Correct for the line-of-sight interstellar extinction for each object using the Q-method for nearby OB stars and relation between $N_{H\,I}$ and E_{B-V} (Humphreys et. al. 2017)
- Derive the stars' luminosities and temperatures during S Dor phases from their (extinction-corrected) M_V and $B-V$.
- Plot the objects on the CMD & HRD and the change in $B-V$ against the change in M_V.



$\Delta(B-V)$ vs. ΔV

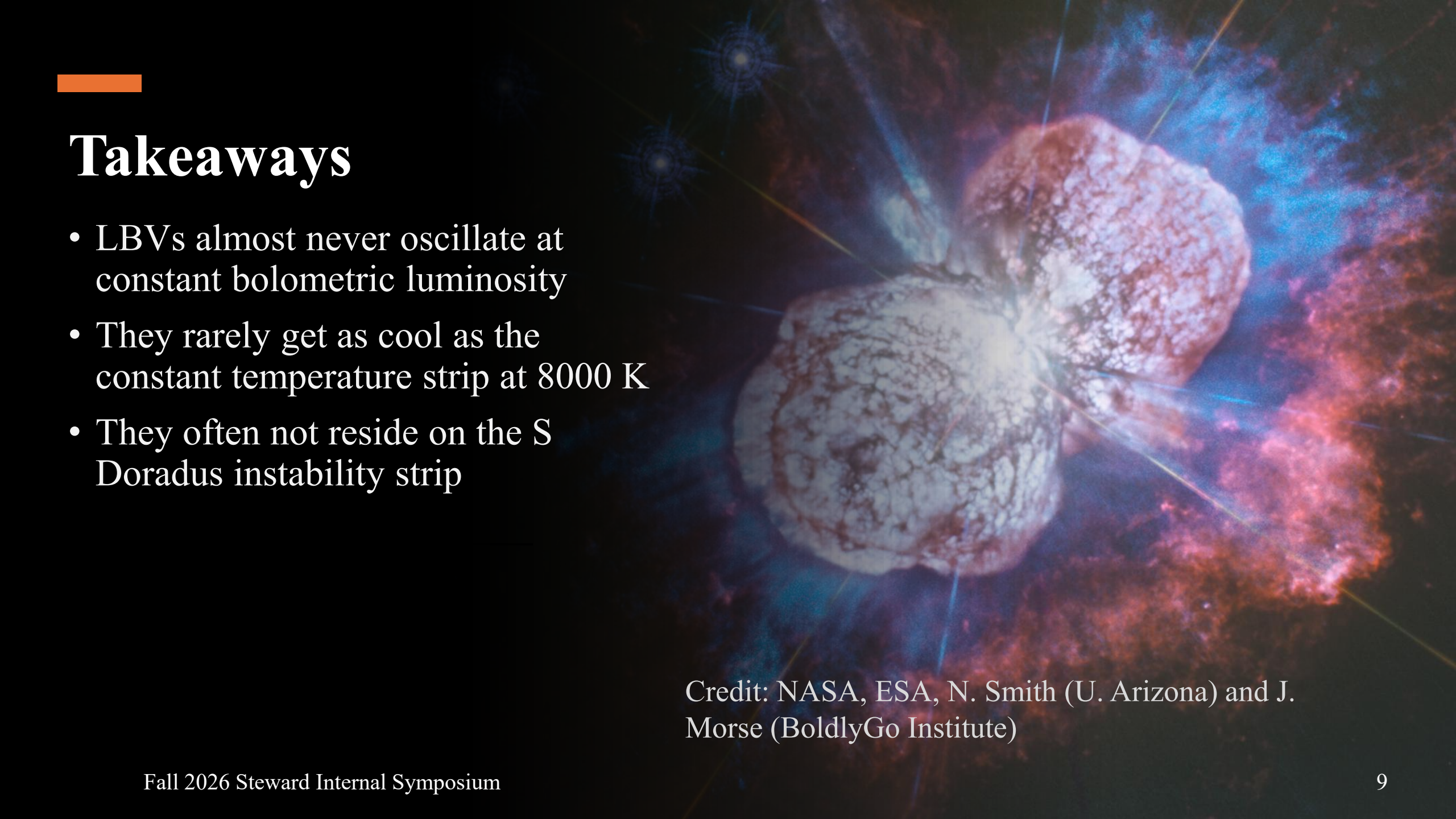


References

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- Smith, N., Milne, P., Bilinski, C., et. al. 2020, MNRAS, 492, 5897-5915
- Szeifert, Th., Humphreys, R. M., Davidson, K., et al. 1996, A&A, 314, 131-145
- Smith, N. 2026, in Encyclopedia of Astrophysics (First Edition), first edition edn., ed. I. Mandel (Oxford: Elsevier), 508–532, doi: https://doi.org/10.1016/B978-0-443-21439-4.00147-4
- Wolf, B. 1989, A&A, 217, 87-91

Takeaways

- LBVs almost never oscillate at constant bolometric luminosity
- They rarely get as cool as the constant temperature strip at 8000 K
- They often not reside on the S Doradus instability strip



Credit: NASA, ESA, N. Smith (U. Arizona) and J. Morse (BoldlyGo Institute)